# WIEM TABBAL

PhD Researcher in Electrical Engineering | Autonomous Underwater Robotics

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## Profile

PhD candidate in Electrical Engineering, specializing in autonomous underwater robotics. I have recently started my doctoral project focusing on object detection and classification in marine environments using computer vision and deep learning.

## Education

**PHD IN ELECTRICAL ENGINEERING**
National Engineering School of Sfax (ENIS)/ Lab-STA (LR11ES50), Tunisia — November 2025 – present
Design and Development of an Autonomous Robotic System for Detection, Classification, and Collection of Marine Waste

- Intelligent perception: Multimodal data fusion and deep learning for robust detection, 3D classification, and discrimination of marine debris.
- Adaptive planning: Development of mission optimization and energy allocation algorithms for multi-robot systems.
- Bio-inspired manipulation: Design and control of a robotic arm for grasping objects of varying shapes and textures in underwater environments.

**ENGINEERING DEGREE IN ELECTRICAL ENGINEERING** (Specialization: Automation and Industrial Computing)
National Engineering School of Sfax (ENIS), Tunisia — September 2020 – June 2023

**PREPARATORY CYCLE IN PHYSICS-CHEMISTRY**
Higher Institute of Applied Sciences and Technology of Gabes, Tunisia — September 2018 – June 2020

## Professional Experience

**PART-TIME LECTURER –** Higher Institute of Technological Studies (ISET) — September 2025 – December 2025

- Facilitated laboratory sessions in **Electrical Circuits** and **Analog Electronics**, bridging the gap between theoretical concepts and hands-on technical applications.
- Managed academic assessments and provided mentorship to students for the successful delivery of their semester-long technical projects.
- Simplified complex technical topics to improve student engagement and proficiency in both circuit analysis software and laboratory hardware.

**ELECTRICAL ENGINEER –** Construction Work Company — July 2024 – July 2025

- Executed comprehensive electrical studies, ensuring strict adherence to safety standards and rigorous technical specifications.
- Produced detailed electrical schematics using **AutoCAD**, optimizing project layouts to streamline the transition from design to execution.
- Directed on-site installations, coordinating efforts between technical teams and contractors to ensure project milestones were met within budget.

**END-OF-STUDIES PROJECT INTERNSHIP –** DRÄXLMAIER SATE — February 2023 – June 2023

- Architected a Full-Stack web application to automate the reporting and tracking of production-line defects, modernizing legacy manual entry systems.

- Developed a responsive frontend using **Angular** and a robust RESTful API with **Spring Boot** to enhance the efficiency of data entry and retrieval.
- Engineered data visualizations by integrating **MySQL** databases with **Power BI**, providing management with real-time insights into manufacturing quality trends.

## Technical Skills

- **AI and Computer Vision:** OpenCV, Keras, TensorFlow
- **Programming:** Python, C/C++, Java
- **Embedded Systems:** Raspberry Pi, STM32, ESP32
- **Tools:** MATLAB/Simulink, LabVIEW
- **Design:** AutoCAD, Proteus
- **Operating Systems:** Linux, Windows

## Academic Projects

**DRIVER DROWSINESS DETECTION SYSTEM** 2023

- Designed a real-time computer vision system using **Python and OpenCV** to monitor driver alertness through facial landmark detection and eye-blink analysis.
- Trained and deployed a **Deep Learning model (Keras/TensorFlow)** on **Raspberry Pi 4**, enabling low-latency inference and real-time safety alerts via hardware peripherals.

**EMBEDDED CONTROL AND IOT SYSTEM WITH RASPBERRY PI** 2023

- Developed a multi-sensor embedded monitoring system using **Raspberry Pi** for real-time data acquisition from the **Pi Camera** and **Sense HAT** sensors.
- Implemented **MQTT-based communication** and built a **ThingSpeak dashboard** for live data visualization and remote monitoring.

**SMART CONNECTED LOCK USING ESP32** 2022

- Engineered a smart access control system using **ESP32 and C++**, enabling secure smartphone-based authentication.
- Developed embedded firmware for wireless communication (Wi-Fi), remote lock management and low-power operation.

## Languages

- French: Fluent
- English: Upper-intermediate to Advanced
- Arabic: Native

## Additional Training and Activities

- STM32 Training – ACTIA Engineering Services (ARM architecture, UART, Embedded C, STM32CubeIDE).
- Member of the organizing committee for the Electrical Engineering Department Open Day (2023)
- Member of the organizing committee for the ENIS ROBOT 4.0 competition (2022)